Systems of many forms with differing degrees

Simon L. Rydin Myerson

10 July 2025

Chalmers University/University of Gothenburg

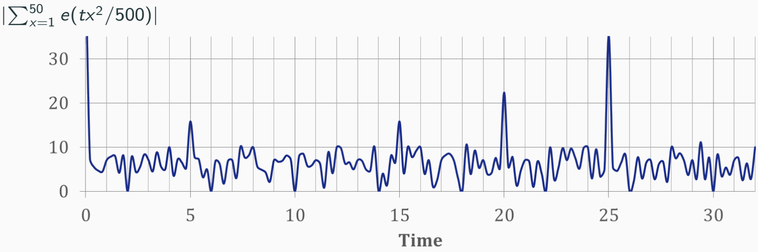
Download these slides:

maths.fan/IML.pdf

Notation

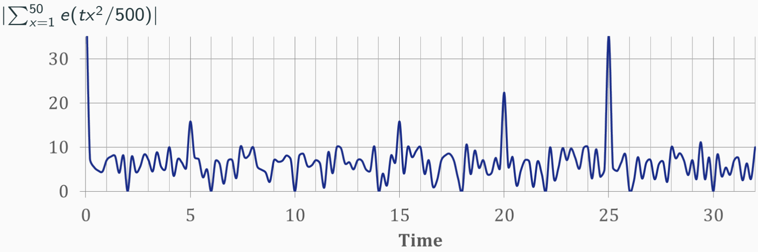
- $\vec{f}(\vec{x}) \in \mathbb{Z}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- Set $N_{\vec{f}}(B) := \#\{\vec{x} \in \mathbb{Z}^s : \vec{f}(\vec{x}) = \vec{0},\ \|\vec{x}\| \leq B\}$, $\vec{\alpha} \cdot \vec{f} = \sum^R \alpha_i f_i$.

$$N_{\vec{f}}(B) = \int_0^1 \cdots \int_0^1 \sum_{\substack{\vec{x} \in \mathbb{Z}^s \\ \|\vec{x}\| \leq B}} e^{2\pi i \vec{t} \cdot \vec{f}(\vec{x})}\, d\vec{t}$$



Repulsion

$$N_{\vec{f}}(B) = \int_0^1 \cdots \int_0^1 \sum_{\substack{\vec{x} \in \mathbb{Z}^s \\ \|\vec{x}\| \leq B}} e^{2\pi i \vec{t} \cdot \vec{f}(\vec{x})} \, d\vec{t}$$



Some peaks bigger than 12 or so. Random noise ≤ 12.

Repulsion: pick points $t, t+\beta$. If both are at peaks, $|\beta| < 1$ or $|\beta| > 4$.

So each peak has width 1, and they are at least 4 apart. Consequently the measure of t lying on peaks is small.

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- We set $N_{\vec{f}}(B) := \#\{\vec{x} \in \mathbb{Z}^s : \vec{f}(\vec{x}) = \vec{0}, \, \|\vec{x}\| \leq B\}$.
- $\vec{f}$ is smooth if $\vec{f} = 0$ defines a smooth $s - R$ dimensional complex manifold in $\mathbb{C}^s \setminus \{\vec{0}\}$(away from the origin).

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1}R(R+1) + R.$$

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- We set $N_{\vec{f}}(B) := \#\{\vec{x} \in \mathbb{Z}^s : \vec{f}(\vec{x}) = \vec{0},\ \|\vec{x}\| \leq B\}$.
- $\vec{f}$ is smooth if $\vec{f} = 0$ defines a smooth $s - R$ dimensional complex manifold in $\mathbb{C}^s \setminus \{\vec{0}\}$(away from the origin).

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1}R(R+1) + R.$$

The Birch-Davenport circle method

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- We set $N_{\vec{f}}(B) := \#\{\vec{x} \in \mathbb{Z}^s : \vec{f}(\vec{x}) = \vec{0},\ \|\vec{x}\| \leq B\}$.
- smooth if $\vec{f} = 0$ defines an $s - R$ dimensional manifold in $\mathbb{C}^s \setminus \{\vec{0}\}$.

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1}R(R+1) + R.$$

The circle method: differing degrees

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1, \ldots, x_s]^R$ is a system of R forms in $s > \sum d_i$ variables with integer coefficients, and $N_{\vec{f}}(B) = \sum_{\vec{x} \in \mathbb{Z}^s \cap [-B,B]^s \setminus \{\vec{0}\}, \vec{f}(\vec{x}) = \vec{0}} 1$.

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1} R(R+1) + R.$$

Generalisations to unequal d_i.

- Browning–Dietmann–Heath-Brown (2014), $d_1 = 2, d_2 = 3$, $s \geq 29$.
- If $d_1, \ldots, d_R \leq d$, Browning–Heath-Brown (2017) handle around $d^3 R^2 2^d$ variables, improving on Schmidt (1985).

The circle method: differing degrees

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1, \ldots, x_s]^R$ is a system of R forms in $s > \sum d_i$ variables with integer coefficients, and $N_{\vec{f}}(B) = \sum_{\vec{x} \in \mathbb{Z}^s \cap [-B,B]^s \setminus \{\vec{0}\}, \vec{f}(\vec{x}) = \vec{0}} 1$.

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1} R(R+1) + R.$$

Generalisations to unequal d_i.

- Browning–Dietmann–Heath-Brown (2014), $d_1 = 2, d_2 = 3$, $s \geq 29$.
- If $d_1, \ldots, d_R \leq d$, Browning–Heath-Brown (2017) handle around $d^3 R^2 2^d$ variables, improving on Schmidt (1985).

Notation

- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1, \dots, x_s]^R$ is a system of R forms in $s > \sum d_i$ variables with integer coefficients, and $N_{\vec{f}}(B) = \sum_{\vec{x} \in \mathbb{Z}^s \cap [-B,B]^s \setminus \{\vec{0}\}, \vec{f}(\vec{x}) = \vec{0}} 1$.

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ smooth, $d_1 = \cdots = d_R = d$,

$$s \geq (d-1)2^{d-1}R(R+1) + R.$$

Generalisations to unequal d_i.

- Browning–Dietmann–Heath-Brown (2014), $d_1 = 2, d_2 = 3$, $s \geq 29$.
- If $d_1, \dots, d_R \leq d$, Browning–Heath-Brown (2017) handle around $d^3 R^2 2^d$ variables, improving on Schmidt (1985).

Quadratic forms with real coefficients

Notation

- $\vec{g}(\vec{x}) \in \mathbb{R}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- The function $M(B) := \#\{\vec{x} \in \mathbb{Z}^s \setminus \{\vec{0}\} : \|\vec{g}(\vec{x})\| \leq 1, \|\vec{x}\| \leq B\}$.
- $\vec{g}$ is nonsingular if the $R \times s$ Jacobian matrix $(\partial g_i(\vec{x})/\partial x_j)_{ij}$ has rank R at every nontrivial complex solution $\vec{x}$ to $\vec{g}(\vec{x}) = \vec{0}$.
- $\vec{g}(\vec{x})$ is irrational if no $\vec{\alpha} \in \mathbb{R}^R \setminus \{\vec{0}\}$ satisfies $\vec{\alpha} \cdot \vec{g}(\vec{x}) \in \mathbb{Z}[\vec{x}]$.

- Bentkus and Götze (1999): circle method proof of Eskin, Margulis and Mozes' asymptotic $M(B) \sim \nu_g B^{s-2}$ in the case when $s \geq 9$.

Quadratic forms with real coefficients

Notation

- $\vec{g}(\vec{x}) \in \mathbb{R}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- The function $M(B) := \#\{\vec{x} \in \mathbb{Z}^s \setminus \{\vec{0}\} : \|\vec{g}(\vec{x})\| \leq 1, \|\vec{x}\| \leq B\}$.
- $\vec{g}$ is nonsingular if the $R \times s$ Jacobian matrix $(\partial g_i(\vec{x})/\partial x_j)_{ij}$ has rank R at every nontrivial complex solution $\vec{x}$ to $\vec{g}(\vec{x}) = \vec{0}$.
- $\vec{g}(\vec{x})$ is irrational if no $\vec{\alpha} \in \mathbb{R}^R \setminus \{\vec{0}\}$ satisfies $\vec{\alpha} \cdot \vec{g}(\vec{x}) \in \mathbb{Z}[\vec{x}]$.

- Bentkus and Götze (1999): circle method proof of Eskin, Margulis and Mozes' asymptotic $M(B) \sim \nu_g B^{s-2}$ in the case when $s \geq 9$.

Theorem (Müller, 2008, slightly rephrased)

If $d_1 = \cdots = d_R = 2$, $\vec{g}$ is nonsingular and irrational, and $s \geq 9R$, then we have $M(B) \sim \nu_{\vec{g}} B^{s-2}$ as $B \to \infty$ for some $\nu \geq 0$.

- Freeman (2000, 2001): Standardised this form of the circle method.
- Buterus-Götze-Hille-Margulis (2022): EMM's $R = 1$, $s \geq 5$, explicit, by circle method! Exp sums $\ll$ nice functions on 1-param groups

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{J}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{J}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Birch 1962, over $\mathbb{Q}$
$(R+1)(d-1)2^{d-2}$
$+R$

"In the style of Birch"

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Birch 1962, over $\mathbb{Q}$
$(R+1)(d-1)2^{d-2}$
$+R$

$\downarrow$

Schmidt 1985,
over $\mathbb{Q}$
$<R^2D^3 2^{3D-5}$

“In the style of Birch”

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

$$\begin{tikzcd}
\begin{array}{c}\text{Birch 1962, over } \mathbb{Q} \\ (R+1)(d-1)2^{d-2} \\ +R\end{array} \arrow[r] \arrow[d] & \begin{array}{c}\text{Skinner 1997} \\ \text{over K} \\ (R+1)(d-1)2^{d-2} \\ +R\end{array} \\
\begin{array}{c}\text{Schmidt 1985,} \\ \text{over } \mathbb{Q} \\ < R^2 D^3 2^{3D-5}\end{array} &
\end{tikzcd}$$

“In the style of Birch”

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

$$
\begin{array}{ccc}
\begin{array}{c}\text{Birch 1962, over } \mathbb{Q} \\ (R+1)(d-1)2^{d-2} \\ +R\end{array} & \longrightarrow & \begin{array}{c}\text{Skinner 1997} \\ \text{over K} \\ (R+1)(d-1)2^{d-2} \\ +R\end{array} \\
\big\downarrow & & \\
\begin{array}{c}\text{Schmidt 1985,} \\ \text{over } \mathbb{Q} \\ <R^2D^3 2^{3D-5}\end{array} & &
\end{array}
$$

Bentkus–
Götze 1999
$R=1, d=2$, over $\mathbb{R}$
9

“In the style of Birch”

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{J}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{J}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Birch 1962, over $\mathbb{Q}$
$(R+1)(d-1)2^{d-2}$
$+R$

$\longrightarrow$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

$\downarrow$

Schmidt 1985,
over $\mathbb{Q}$
$<R^2 D^3 2^{3D-5}$

Bentkus–
Götze 1999,
Müller 2005
$d=2$, over $\mathbb{R}$
$9R$

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{J}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{J}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–Heath-Brown 2009, Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Birch 1962, over $\mathbb{Q}$
$(R+1)(d-1)2^{d-2}+R$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}+R$

Schmidt 1985,
over $\mathbb{Q}$
$<R^2D^3 2^{3D-5}$

Bentkus–Götze 1999,
Müller 2005
d=2, over $\mathbb{R}$
$9R$

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{J}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{J}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–Heath-Brown 2009, Hanselmann 2012
d=4 over $\mathbb{Q}$
40

$\longleftarrow$

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

$\longleftarrow$

Birch 1962, over $\mathbb{Q}$
$(R+1)(d-1)2^{d-2}+R$

$\longrightarrow$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}+R$

$\downarrow$

Schmidt 1985,
over $\mathbb{Q}$
$<R^2D^3 2^{3D-5}$

Bentkus–Götze 1999,
Müller 2005
d=2, over $\mathbb{R}$
$9R$

"In the style of Birch"

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–
Heath-Brown 2009,
Hanselmann 2012
$d{=}4$ over $\mathbb{Q}$
40

Davenport 1963
Heath-Brown 2007
$R{=}1, d{=}3$, over $\mathbb{Q}$
(nonsingular-ish)
14

Birch 1962, over $\mathbb{Q}$
Dietmann 2015,
Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}$
$+R$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

Browning–
Prendiville 2017
$R{=}1$, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

Schmidt 1985,
over $\mathbb{Q}$
$<R^2D^3 2^{3D-5}$

Bentkus–
Götze 1999,
Müller 2005
$d{=}2$, over $\mathbb{R}$
$9R$

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–Heath-Brown 2009, Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Birch 1962, over $\mathbb{Q}$
Dietmann 2015, Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}+R$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}+R$

Browning–Prendiville 2017
R=1, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

Schmidt 1985, Browning–Heath-Brown 2017, over $\mathbb{Q}$
$<R^2D^22^{D-1}$

Bentkus–Götze 1999, Müller 2005
d=2, over $\mathbb{R}$
$9R$

“In the style of Birch”

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–Heath-Brown 2009, Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Birch 1962, over $\mathbb{Q}$
Dietmann 2015,
Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}$
$+R$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

Browning–Prendiville 2017
R=1, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

Schmidt 1985,
Browning–Heath-Brown
2017, over $\mathbb{Q}$
$<R^2D^22^{D-1}$

Madritsch–Frei
2017, over K
$<R^2D^22^{D-1}$

Bentkus–Götze 1999,
Müller 2005
d=2, over $\mathbb{R}$
$9R$

“In the style of Birch”

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Birch 1962, over $\mathbb{Q}$
Dietmann 2015,
Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}$
$+R$

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Browning–
Heath-Brown 2009,
Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Browning–
Prendiville 2017
R=1, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

Schmidt 1985,
Browning–
Heath-Brown
2017, over $\mathbb{Q}$
$<R^2D^22^{D-1}$

Madritsch–Frei
2017, over K
$<R^2D^22^{D-1}$

Bentkus–
Götze 1999,
Müller 2005
d=2, over $\mathbb{R}$
9R

RM 2017, 2018
d=2, over $\mathbb{Q}$
9R

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\leq 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Birch 1962, over $\mathbb{Q}$
Dietmann 2015,
Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}$
$+R$

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Browning–
Heath-Brown 2009,
Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

Browning–
Prendiville 2017
R=1, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

RM 2017/2019
d=3, $\mathbb{R}$ or $\mathbb{Q}$
$25R$

Schmidt 1985,
Browning–
Heath-Brown
2017, over $\mathbb{Q}$
$<R^2D^22^{D-1}$

Madritsch–Frei
2017, over K
$<R^2D^22^{D-1}$

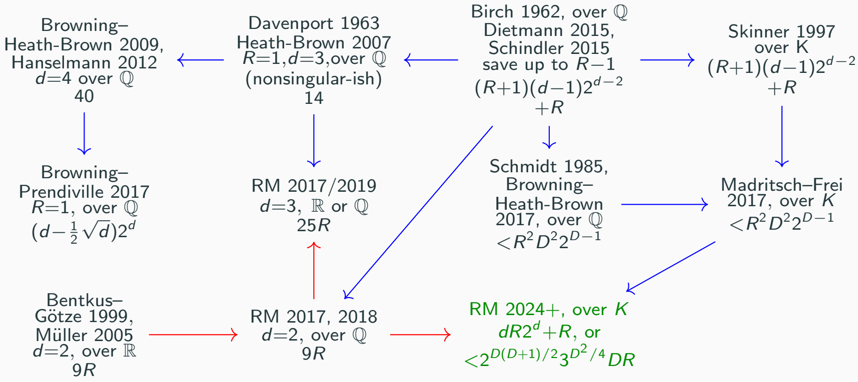
Bentkus–
Götze 1999,
Müller 2005
d=2, over $\mathbb{R}$
$9R$

RM 2017, 2018
d=2, over $\mathbb{Q}$
$9R$

"In the style of Birch"

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{J}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{J}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.



“In the style of Birch”

maths.fan/IML.pdf

The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \cdots = d_R = d$ or $2 \le d_i \le D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. No averaging over $\vec{a}$ with $\vec{\alpha} \approx \vec{a}/q$.
- Show $\sum_{\vec{x}:\vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathfrak{S}\mathfrak{I}B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K, or $\sum_{\vec{x}:|\vec{f}|\le 1/2} w(\vec{x}/B) \sim \mathfrak{I}P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s.

Browning–Heath-Brown 2009, Hanselmann 2012
d=4 over $\mathbb{Q}$
40

Davenport 1963
Heath-Brown 2007
R=1,d=3,over $\mathbb{Q}$
(nonsingular-ish)
14

Birch 1962, over $\mathbb{Q}$
Dietmann 2015, Schindler 2015
save up to $R-1$
$(R+1)(d-1)2^{d-2}$
$+R$

Skinner 1997
over K
$(R+1)(d-1)2^{d-2}$
$+R$

Browning–Prendiville 2017
R=1, over $\mathbb{Q}$
$(d-\frac{1}{2}\sqrt{d})2^d$

RM 2024+, over $\mathbb{R}$
$d(d-3)R2^{2+d}+R$, or
$<2^{2+D(D+1)/2}3^{D^2/4}D^2R$

Schmidt 1985, Browning–Heath-Brown 2017, over $\mathbb{Q}$
$<R^2D^22^{D-1}$

Madritsch–Frei 2017, over K
$<R^2D^22^{D-1}$

Bentkus–Götze 1999, Müller 2005
d=2, over $\mathbb{R}$
$9R$

RM 2024+, over K
$dR2^d+R$, or
$<2^{D(D+1)/2}3^{D^2/4}DR$

Theorem (RM 2024+)

$N_{\vec{f}}(B) \sim c_{\vec{f}} B^{m(s-|\vec{d}|)}$ if $\vec{f} = 0$ is smooth, and, for $R_d = \#\{i : d_i \geq d\}$,

$$\frac{(s+1-R_D)\min\{1, 2RD/(R+1)(D-1)\}}{2^D} + \sum_d \frac{(s+1-R_d)\min\{1, 2Rd/(R+1)(d-1)\}}{(2^{D-1}+1)\cdots(2^d+1)2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|.$$

Typically this is about $2^{D(D+1)/2} 3^{d^2/4} DR$ variables; when $d_1 = \cdots = d_R = d$ it is $(1 + d2^d)R$ variables.

Ideas: $\mathfrak{p}$-adic repulsion ($g \not\equiv h \bmod \mathcal{O}_K[\vec{x}] \Rightarrow |\sum e(f(\vec{x})) \sum e(g(\vec{x}))| < |g - h|_{\mathfrak{p}}^C$), accessing low degrees in $\sum e(f(\vec{x}))$ ("reiterated differencing").

Theorem (RM 2024+)

$N_{\vec{f}}(B) \sim c_{\vec{f}} B^{m(s-|\vec{d}|)}$ if $\vec{f}=0$ is smooth, and, for $R_d = \#\{i : d_i \geq d\}$,

$$\frac{(s+1-R_D)\min\{1,2RD/(R+1)(D-1)\}}{2^D} + \sum_d \frac{(s+1-R_d)\min\{1,2Rd/(R+1)(d-1)\}}{(2^{D-1}+1)\cdots(2^d+1)2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|.$$

Typically this is about $2^{D(D+1)/2} 3^{d^2/4} DR$ variables; when $d_1 = \cdots = d_R = d$ it is $(1+d2^d)R$ variables.

Ideas: $\mathfrak{p}$-adic repulsion ($g \not\equiv h \bmod \mathcal{O}_K[\vec{x}] \Rightarrow |\sum e(f(\vec{x})) \sum e(g(\vec{x}))| < |g-h|_{\mathfrak{p}}^C$), accessing low degrees in $\sum e(f(\vec{x}))$ ("reiterated differencing").

Theorem (RM 2024+)

$s'_d = s - R_d$ ($d \leq 3$) or $s_d^{(1)} = [\frac{s+1-R_d}{2(d-3)} + \frac{1}{2}]$, $s'_d = \frac{s_d^{(1)}}{2}(1 - \frac{(d-3)(s_d^{(1)}-1)}{s+1-R_d})$ ($d \geq 4$). Suppose that $\vec{g}$ is nonsingular and irrational, and that

$$\frac{s'_D}{2^D} + \sum_d \frac{s'_d}{(2^{D-1}+1)\cdots(2^d+1)2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|$$

Then $M(B) \sim \nu_{\vec{g}} B^{n-\sum d_i}$ where $\nu_{\vec{g}} = \lim B^{\sum d_i} \mathbb{P}[\vec{g}(\vec{X}) \in [0,1)^R]$.

$s'_d \approx s/3(d-3)$; about $8(D-3)$ times as many variables. Ideas: reiterated differencing, work of Davenport, very pedantic book-keeping.

Setup for repulsion

- $e(t) = 2^{2\pi i t}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1,\dots,\vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f, with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}^{f,k}_{\vec{h}_1,\dots,\vec{h}_{k-1}}$ is the vector of coefficients of $\Delta_{\vec{h}_1,\dots,\vec{h}_{k-1},(\cdot)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x} \in \mathbb{Z}^s} B^{-s} w(\vec{x}/B) e\left(f(\vec{x})\right)$

For $f, g \in \mathbb{R}[\vec{x}]$ with degree $\le k$, and $pB^{-k} \ll |f^{[k]}| \ll p^{1-k}$ we have

$$|S_w(B; f) S_w(B; g + f)|^{2^{k-1}} \ll p^{-(k-1)s}$$
$$\cdot \#\{(\vec{h}_1, \dots, \vec{h}_{k-1}) \in \mathbb{Z}^{(k-1)s} : |\vec{h}_i| \le p, |\vec{m}^{f,k}_{\vec{h}_1,\dots,\vec{h}_{k-1}}| \le p^{k-2} |f^{[k]}|\}.$$

If $k = 3$, $f^{[3]}$ smooth get $\ll p^{-s}$ by ideas of Davenport; else, pigeonhole.

Repulsion: suppose that, whenever $pB^{-k} \le |\vec{\beta} \cdot \vec{f}^{[k]}| \le p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w\left(B; (\vec{\alpha} + \vec{\beta}\right) \cdot \vec{f}) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0,1)^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i (A/E(d_i) - 1)}$ $(A > 0)$

$\bigcirc$: If $\sum^R d_i / E(d_i) < 1$ then $\begin{cases} N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s - \sum^R d_i} & \text{if } \vec{f} \in \mathbb{Z}[\vec{x}]^R, \text{ or} \\ M(B) \sim \nu_{\vec{f}} B^{s - \sum^R d_i} & \text{if } \vec{f} \text{ is irrational.} \end{cases}$

Setup for p-adic repulsion

- $e(t) = 2^{2\pi i t}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1,\ldots,\vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f, with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}^{f,k}_{\vec{h}_1,\ldots,\vec{h}_{k-1}}$ is the vector of coefficients of $\Delta_{\vec{h}_1,\ldots,\vec{h}_{k-1},(\,\cdot\,)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x}\in\mathbb{Z}^s} B^{-s} w(\vec{x}/B) e\left(f(\vec{x})\right)$

For $f \in \mathbb{Z}[\vec{x}], g \in \mathbb{R}[\vec{x}]$ with degree $\leq k$, and p prime with $p^M \ll B^k$, such that $p^{-M} < |f^{[k]}|_p \leq p^{1-k}$ we have

$$|S_w(B; f) S_w(B; g + \tfrac{1}{p^M} f)|^{2^{k-1}} \ll p^{-(k-1)s}$$
$$\cdot \#\{(\vec{h}_1, \ldots, \vec{h}_{k-1}) \in \mathbb{Z}^{(k-1)s} : |\vec{h}_i| \leq p, |\vec{m}^{f,k}_{\vec{h}_1,\ldots,\vec{h}_{k-1}}|_p \leq p^{-1} |f^{[k]}|_p\}.$$

For $f^{[k]}$ smooth this is $\ll p^{-s}$ ($\mathbb{F}_p$-points on varieties).

Repulsion: suppose that, whenever $p^{-M} < |\vec{b} \cdot \vec{f}^{[k]}|_p \leq p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w\left(B; (\vec{\alpha} + \tfrac{\vec{b}}{p^M}) \cdot \vec{f}\right) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0,1)^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i (A/E(d_i) - 1)}$ $(A > 0)$

◯: If $\sum^R d_i / E(d_i) < 1$ then $N_{\vec{f}}(B) \sim c B^{s - \sum^R d_i}$. Smooth: $E(d) = \frac{n-R+1}{2^d}$.

Measures by repulsion

- $e(t) = 2^{2\pi i t}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x}+\vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1,\dots,\vec{h}_r} = \Delta_{\vec{h}_1}\cdots\Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f, with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}^{f,k}_{\vec{h}_1,\dots,\vec{h}_{k-1}}$ is the vector of coefficients of $\Delta_{\vec{h}_1,\dots,\vec{h}_{k-1},(\,\cdot\,)} f^{[k]}$
- The *normalised* sum $S_w(B;f) = \sum_{\vec{x}\in\mathbb{Z}^s} B^{-s} w(\vec{x}/B) e\left(f(\vec{x})\right)$
- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1,\dots,x_s]^R$ is a system of R forms in $s > \sum d_i$ variables.

Repulsion: suppose that, whenever $pB^{-k} \le |\vec{\beta}\cdot\vec{f}^{[k]}| \le p^{1-k}$, we have

$$S_w(B;\vec{\alpha}\cdot\vec{f}) S_w\left(B;(\vec{\alpha}+\vec{\beta}\right)\cdot\vec{f}) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha}\in[0,1)^R : |S_w(B;\vec{\alpha}\cdot\vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i(A/E(d_i)-1)}$ $(A>0)$

proof idea: Let $p \gg B^{A/E(k)}$. Let $\vec{\alpha}, \vec{\alpha}+\frac{\vec{b}}{p^M}$ belong to the set. Then

$$|\vec{\beta}\cdot\vec{f}^{[k]}| < pB^{-k}, \text{ or } |\vec{\beta}\cdot\vec{f}^{[k]}| > p^{1-k}.$$

It follows that the $\vec{\alpha}\cdot\vec{f}^{[k]}$ are contained in a few infrequent regions (peaks) of diameter $\le pB^{-k}$, separated by gaps of size $\ge p^{1-k}$, hence with total measure $\le (p/B)^{kR_k}$ where there are R_k forms of degree k.

Measures by p-adic repulsion

- $e(t) = 2^{2\pi i t}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x}+\vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1,\dots,\vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f, with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}^{f,k}_{\vec{h}_1,\dots,\vec{h}_{k-1}}$ is the vector of coefficients of $\Delta_{\vec{h}_1,\dots,\vec{h}_{k-1},(\,\cdot\,)} f^{[k]}$
- The *normalised* sum $S_w(B;f) = \sum_{\vec{x}\in\mathbb{Z}^s} B^{-s} w(\vec{x}/B) e\left(f(\vec{x})\right)$
- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1,\dots,x_s]^R$ is a system of R forms in $s > \sum d_i$ variables.

Repulsion: suppose that, whenever $p^{-M} < |\vec{b}\cdot\vec{f}^{[k]}|_p \leq p^{1-k}$, we have

$$S_w(B;\vec{\alpha}\cdot\vec{f}) S_w\left(B;(\vec{\alpha}+\tfrac{\vec{b}}{p^M})\right)\cdot\vec{f}) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha}\in[0,1)^R : |S_w(B;\vec{\alpha}\cdot\vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i(A/E(d_i)-1)}$ $(A>0)$

proof idea: Let $p \gg B^{A/E(k)}$ prime, $p^M \ll B^k$. Let $\vec{\alpha}, \vec{\alpha}+\frac{\vec{b}}{p^M}$ belong to the set. Either $\vec{\beta}=0$ or $\vec{\beta}\in p^{k-2-M}\mathbb{Z}$, because

$$|\vec{b}\cdot\vec{f}^{[k]}|_p \leq p^{-M}, \text{ or } |\vec{b}\cdot\vec{f}^{[k]}|_p > p^{1-k}.$$

Hence each lattice $\vec{\alpha}_0 + \frac{1}{p^{k-1-M}}\mathbb{Z}^R$ contains at most one element of the set, hence it has total measure $\leq p^{(k-1-M)R_k}$ where there are R_k forms of degree k, this is $\leq (p/B)^{kR_k}$ if we choose M so $B^{-k} \gg p^{-M-1}$.

Notation

- $e(t) = 2^{2\pi i t}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1,\ldots,\vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- NB $\Delta_{\vec{h}_1,\ldots,\vec{h}_k} f^{[k]}$ is a multilinear form in the $\vec{h}_i$, independent of $\vec{x}$
- $\vec{m}^{f,k}_{\vec{h}_1,\ldots,\vec{h}_{k-1}}$ is the vector of coefficients of $\Delta_{\vec{h}_1,\ldots,\vec{h}_{k-1},(\,\cdot\,)} f^{[k]}$

Proposition

Given $w \in C_c^\infty(\mathbb{R}^s)$ there is $\tilde{w} \in C_c^\infty(\mathbb{R}^{ks})$ as follows. For any $f \in \mathbb{R}[\vec{x}]$ of degree $\leq d$, and any $1 < k < d$,

$$\left|\sum_{\vec{x}\in\mathbb{Z}^s} \frac{1}{B^s} w\left(\frac{\vec{x}}{B}\right) e\left(f(\vec{x})\right)\right|^{(2^{d-1}+1)\cdots(2^k+1)2^{k-1}3^{(k+1)(d-k-1)+1}}$$

$$\leq B^{-ks} \sum_{\vec{h}_1,\ldots,\vec{h}_{k-1}\in\mathbb{Z}^s} \left|\sum_{\vec{h}_k\in\mathbb{Z}^s} \tilde{w}\left(\frac{\vec{h}_1}{B},\ldots,\frac{\vec{h}_k}{B}\right) e\left(\Delta_{\vec{h}_1,\ldots,\vec{h}_k} f^{[k]}\right)\right|$$

$$\ll \frac{1}{B^{(k-1)s}} \#\{(\vec{h}_1,\ldots,\vec{h}_{k-1},\vec{u}) \in \mathbb{Z}^{ks} : |\vec{h}_i| \leq B, |\vec{m}^{f,k}_{\vec{h}_1,\ldots,\vec{h}_{k-1}} - \vec{u}| \leq \frac{1}{B}\}$$

Lemma

Given $w \in C_c^\infty(\mathbb{R}^s)$ there is $\tilde{w} \in C_c^\infty(\mathbb{R}^{(d+1)s})$ as follows. For any $f \in \mathbb{R}[\vec{x}]$ of degree $\leq d$,

$$\left|\sum_{\vec{x}\in\mathbb{Z}^s} \frac{1}{B^s} w\left(\frac{\vec{x}}{B}\right) e\left(f(\vec{x})\right)\right|^{2^{d-1}+1}$$
$$\leq \left| \sum_{\vec{x}_1,\ldots,\vec{x}_d} \sum_{\vec{x}} \tilde{w}\left(\frac{\vec{x}_d}{B},\ldots,\frac{\vec{x}_1}{B},\frac{\vec{x}}{B}\right) e\left(f^{[<d]} + F\left(\vec{x};\vec{x}_1,\ldots,\vec{x}_d\right)\right) \right|$$

with $F = f^{[d]} - \Delta_{\vec{x}+\vec{x}_1,\ldots,\vec{x}+\vec{x}_d} f^{[d]}$ of degree $< d$ in $\vec{x}$ and ≤ 1 in each $\vec{x}_i$.

Lemma

For $L \in \mathrm{GL}_s(\mathbb{R})$, $A := \{\|\vec{x}\| \leq B : \|L\vec{x} - \vec{u}\| \leq 1/B \text{ some } \vec{u} \in \mathbb{Z}^s\}$. Given $w \in C_c^\infty(\mathbb{R}^{2s})$ there are $\tilde{w}_{\vec{c}}^{L,B} \in C_c^\infty(\mathbb{R}^s)$, whose values, support and derivatives are bounded in terms of w only, such that for all real g,

$$\left|\sum_{\vec{x},\vec{y}} w\left(\frac{\vec{x}}{B},\frac{\vec{y}}{B}\right) e\left(\vec{y}\cdot L\vec{x} + g(\vec{x})\right)\right|^3 \leq \sum_{\vec{c}\in A-A} \sum_{\vec{x}} \tilde{w}_{\vec{c}}^{L,B}\left(\frac{\vec{x}}{B}\right) e\left(\Delta_{\vec{c}} g(\vec{x})\right).$$